

Intra-Plane Layer $R/T(p)$ Projection in the Thesis Symplectic Energy Normalisation (IntraPlaneFoldyLax Phase 3a)

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1 Goal

Project the Phase-2 planar collective scattering operator $\mathbf{T}_{\text{coll}}(\mathbf{k}_{\parallel})$ onto up/down P - SV - SH plane waves at horizontal slowness p , producing the layer reflection/ transmission operator $\{\mathbf{R}_D, \mathbf{R}_U, \mathbf{T}_D, \mathbf{T}_U\}$ (2×2 P - SV) and the SH scalar, across normal, sub-critical, and post-critical incidence. The energy normalisation is the one defined in Section 3.1 of the thesis (the free-space elastic Green's tensor), *not* the codebase's velocity-weighted `slab.to_modified` convention.

2 Energy normalisation (thesis Section 3.1)

The Fourier-domain system matrix $\mathbf{A}(k_x, k_y)$ is quasi-Hamiltonian, $\mathbf{A}^T(-\mathbf{k})\mathbf{J}_6 + \mathbf{J}_6\mathbf{A}(\mathbf{k}) = \mathbf{0}$. Its eigenvectors are the down(+)/up(-) $P/SV/SH$ plane waves $|\pm, c, \mathbf{k}\rangle$ (thesis Eqs. Peigen/SVeigen/SHeigen), each a six-vector (displacement, traction) times a normalisation factor ϵ_c chosen (Eq. epsdef) so that the symplectic eigenvector product is unit,

$$(\mathbf{J}_6 \mathbf{D}_z(-\mathbf{k}))^T \mathbf{D}_z(\mathbf{k}) = i \mathbf{J}_6. \quad (1)$$

At $k_y = 0$, with $k_{z,c} = \omega \eta_c$ (η_c the vertical slowness, $\text{Im } \eta_c > 0$ evanescent) and $\hat{K}_S = \hat{K}_H = \omega/\beta$,

$$\epsilon_P = \frac{1}{\sqrt{2\rho\omega^2 k_{z,P}}}, \quad \epsilon_S = \frac{\omega}{\beta \hat{K}_S \sqrt{2\rho\omega^2 k_{z,S}}}, \quad \epsilon_H = \frac{1}{\hat{K}_H \sqrt{2\rho\omega^2 k_{z,H}}}. \quad (2)$$

3 Projection pipeline

Per slowness p and incident mode m :

1. **Incident.** The incident wave is the ϵ -normalised down-going eigenvector $|+, m\rangle$ (direction $\hat{\mathbf{e}}_m$, complex scale $\mu_m = \hat{\mathbf{e}}_m \cdot \mathbf{u}_m^E$), expanded into regular multipoles via the verified incident bridge. For an evanescent ($\text{Im } \eta_m > 0$) direction the bridge uses the analytic continuation $\overline{Y_n^m(\hat{\mathbf{k}})} = (-1)^m Y_n^{-m}(\hat{\mathbf{k}})$ (correct for complex $\hat{\mathbf{k}}$; naïvely conjugating the direction is wrong and was the post-critical bug). A transverse incident wave, SV or SH , carries *both* $\mathbf{N}$ and $\mathbf{M}$ multipoles whatever its polarisation; keeping only $\mathbf{N}$ for SV and only $\mathbf{M}$ for SH breaks energy balance and leaves a reciprocity residual of 10^{-10} – 10^{-8} .
2. **Scatter.** $\mathbf{b} = \mathbf{T}_{\text{coll}} \mathbf{a}$, with $\mathbf{T}_{\text{coll}} = \mathbf{T}_0(\mathbf{I} - \mathbf{G}_0 \mathbf{T}_0)^{-1}$ at Bloch vector $\mathbf{k}_{\parallel} = \omega(0, p, 0)$.

3. **Project.** The specular scattered displacement at the up/down direction of mode c , $\mathbf{U}^{\text{sc}} = (2\pi i/\eta_c \omega A_{\text{cell}}) \mathbf{f}$ (the lattice Weyl prefactor times the multipole far-field $\mathbf{f}$, with $\mathbf{u} \sim \mathbf{f} e^{ikr}/r$: summed over the lattice, e^{ikr}/r is $2\pi i/(A_{\text{cell}} k_z)$ per order), is projected onto the ϵ -normalised $|\text{out}, c\rangle$: $R_c^E = (\hat{\mathbf{e}}_c \cdot \mathbf{U}^{\text{sc}})/\mu_c$.

No separate flux matrix $\mathbf{D}$ and no fitted constant appear; the ϵ_c carry the energy normalisation.

4 Result: symplectic reciprocity at all p

In the thesis ϵ -energy normalisation the reciprocity is the *symplectic* form: reflection is antisymmetric and transmission obeys a parity relation with $\mathbf{\Sigma} = \text{diag}(1, -1)$ (SV odd),

$$\mathbf{R}_D = -\mathbf{R}_D^T, \quad \mathbf{R}_U = -\mathbf{R}_U^T, \quad \mathbf{T}_U = \mathbf{\Sigma} \mathbf{T}_D \mathbf{\Sigma}. \quad (3)$$

These hold to machine/quadrature precision at every p , including post-critical (where P is evanescent):

regime	$\ \mathbf{R}_D + \mathbf{R}_D^T\ $	$\ \mathbf{R}_U + \mathbf{R}_U^T\ $	$\ \mathbf{T}_U - \mathbf{\Sigma} \mathbf{T}_D \mathbf{\Sigma}\ $
normal ($p = 10^{-6}$)	4.7×10^{-19}	4.9×10^{-18}	1.2×10^{-17}
sub-critical ($p = 0.5/\alpha$)	2.5×10^{-16}	2.6×10^{-16}	2.6×10^{-18}
post-critical ($p = 0.8/\beta$)	3.2×10^{-16}	3.2×10^{-16}	2.3×10^{-18}

All three are at round-off. The reflection residuals of 10^{-10} – 10^{-8} previously reported here were not a quadrature floor: they came from the incomplete incident expansion above. The plain-symmetric $\mathbf{T}_U = \mathbf{T}_D^T$, $\mathbf{R}_D$ symmetric is the codebase-Kennett (velocity-weighted) convention, which differs from the thesis by the factor i on SV ; that factor converts symmetric $\leftrightarrow$ antisymmetric. Both are valid; the thesis-native form is symplectic.

5 Scope and deferral

Phase 3a establishes the projection and its reciprocity across the full slowness range. The lossless energy balance $\|R\|^2 + \|T\|^2 = 1$ is deferred to Phase 3b: it requires the *undamped* vector $\mathbf{G}_0$ (Ewald/Kambe), because the Phase-2 lattice damping $\text{Im } \kappa = 0.25$ is an artificial loss that preserves reciprocity but breaks energy conservation. Numerical settings: $ka = 0.3$ (sub-wavelength, single propagating Bragg order at pitch $a_L = 2.5$), multipole order $N = 3$, lattice half-width $L = 4$; the moderate test contrast ($\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$).

Deliverables: `Mathematica/IntraPlaneRT.wl` (+ `.nb` twin), `IntraPlaneRT_reference.json`, and the Python cross-check `cubic_scattering/tests/test_intraplane_rt.py`.